Original Article

Artificial intelligence models cannot yet replace experts in providing patient education for shoulder and elbow orthopaedic pathologies: a systematic review and meta-analysis

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Background: AI and machine learning (ML) have diverse applications in orthopedic surgery, such as for diagnosis of disease, surgical assistance, and outcome prediction. When used as adjuncts, AI has a potential to reduce clinical workload, improve workflow and aid in clinical decision making. The objective of this systematic review is to evaluate current literature on artificial intelligence (AI) to assess effectiveness in developing responses to inquiries related to orthopaedic upper extremity pathologies.

Methods: Three databases (PubMed, MEDLINE, EMBASE) were searched for studies involving AI and questions in shoulder and elbow orthopedics. Inclusion criteria included papers related to shoulder and elbow, human studies, use of AI models, published in English language and at any level of evidence. Data on response accuracy, reliability and quality, as well as area under the curve (AUC) of the given AI algorithm, were recorded. Meta-analyses were conducted on both the accuracy and AUC of AI algorithms on relevant studies. Risk of bias was assessed using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2).

Results: A total of 16 studies were included in this review. Nine studies used a version of ChatGPT, one study used GoogleBard, and the remaining seven studies used a variety of AI learning models. The overall pooled accuracy of responses developed by AI models was 78%, and the pooled mean AUC of included AI algorithms was 86%. AI algorithms performed inferiorly compared to experts. The overall quality and readability of AI responses were poor.

Conclusions: AI algorithms assessed in our study demonstrated a promising degree of accuracy and performance. However, AI responses were found to be inferior to experts and had poor readability, quality, and value to the patient. In its current state of technology, AI is a powerful tool that can be used in conjunction with experts to augment patient education, however, it should not be utilized independently.

Keywords: Artificial intelligence (AI); large language model (LLM); patient education; upper extremity

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Introduction

With recent major advances in artificial intelligence (AI) and machine learning (ML), there is significant potential to transform patient care and improve healthcare outcomes. The applications of AI in the field of orthopaedic surgery are diverse; for example, identification of pathologies or issues with implants on radiographs, assisting in doctor-patient communication and documentation, predicting clinical outcomes and length of stay, and assisting in real-time surgeries, among many others (1,2). While these applications are seemingly endless, particular interest has been garnered regarding the use of large language models (LLMs) in improving accessibility and quality of patient education. A LLM is a deep-learning algorithm, pre-trained with massive datasets and parameters, that uses natural language processing to manage complex tasks (1,3). OpenAI's Generative Pre-Trained Transformer (ChatGPT), one of the most popular chatbots, has recently seen rapid growth and development (2). Since the advent of ChatGPT, many other LLMs and ML models have been developed and made publicly accessible, such as Google's Gemini (formerly known as Bard), Microsoft's Copilot and XGBoost.

The ability of LLMs to identify a user's question, evaluate and integrate vast amounts of information from diverse sources and translate it into a digestible and conversational response demonstrates a promising future for AI in patient education. AI models have been shown to pass the United States Medical Licensing Examination (USMLE) board exams with scores at or above the passing threshold of 60% (4-6). Furthermore, despite demonstrating inferior performance compared to humans, ChatGPT was found to be effective in answering specialty-specific information on the American Shoulder and Elbow Surgeons (ASES) maintenance of certification exam (7). AI is already extensively being used in the field of orthopaedic surgery, including in joint replacements, automated image-based diagnosis, and predicting clinical outcomes (8). When AI algorithms are utilized as augmentation tools by experts, reported findings demonstrate reduced clinician workload, more efficient workflow, and improved shared decision-making with patients, among many other benefits (9-12). However, concern has been raised over the consistent accuracy of ChatGPT's responses to text inputs; it has been reported to provide incorrect and outdated information and even sometimes 'hallucinate' by generating very realistic yet fake information (2,13,14).

There is yet to be an extensive assessment of the effectiveness and reliability of AI in shoulder and elbow orthopaedic surgery. However, this is a rapidly developing field of research. For example, a recent cross-sectional study provided a comparison of the quality of responses between ChatGPT, Bing Chat, and AskeOE AI chatbots to a range of treatment-related questions across various orthopaedic surgery subspecialties (15). To understand whether AI will be effective in educating patients on orthopaedic surgery-related topics, it is important to first establish whether responses developed by AI chatbots are consistently accurate, reliable, and easy to understand. If AI is empirically shown to achieve these goals, it may save considerable time and resources for both the surgeon and the patient. This paper is the first to systematically assess how AI can impact patient education in the field of shoulder and elbow orthopaedic surgery and will help to

Highlight box

Key findings

- Artificial intelligence (AI) demonstrated promising accuracy, but the answers produced were inferior to experts in readability, quality, and patient value.
- As AI's role continues to be defined within the field of orthopedics, this study suggests that it may serve as a useful adjunct, but its independent use should be limited.

What is known and what is new?

- AI and machine learning have diverse applications in orthopedic surgery such as for diagnosis of disease, surgical assistance, and outcome prediction. When used as adjuncts, AI has the potential to reduce clinical workload, improve workflow and aid in clinical decision making.
- This study was the first to assess the role of AI in evaluating the reliability, accuracy, and readability of shoulder and elbow surgery patient education material, finding that AI's role is useful as an adjunct tool to clinician expertise, rather than as an independent resource.

What is the implication and what should change now?

- AI models demonstrate promising accuracy; however, their educational responses lack readability, quality, and patient-centered value. The current AI responses are above most patients' comprehension potential, which limits use in patient education.
- The technology cannot yet replace expert opinion due to the presence of poorer readability, the absence of quality standards, and variability.
- Future directions include understanding which questions patients value most to shape AI model development, refining AI output algorithms that meet standards of accuracy and readability, and developing guidelines for safe AI use in patient education, as well as training physicians to safely integrate AI into care.

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guide decision-making regarding routinely implementing AI in patient education and care. The authors hypothesized that AI models would produce responses with similar accuracy yet lower readability scores compared to humans. The article follows the Revised Assessment of Multiple Systematic Reviews (R-AMSTAR) guidelines (16). We present this article in accordance with the PRISMA reporting checklist (available at https://aoj.amegroups.com/article/view/10.21037/aoj-25-38/rc) (17).

Methods

Search criteria

Three online databases (PubMed, MEDLINE, EMBASE) were searched from database inception to June 6, 2024, to identify studies that assessed the accuracy of AI in patient education regarding shoulder and elbow procedures or pathologies. Comprehensive search terms including 'artificial intelligence, 'upper extremity', 'shoulder', 'elbow', 'clavicle', 'scapula', 'humerus', 'total shoulder replacement', 'rotator cuff tear', 'arthroscopy', and 'dislocation' were used (Table S1). The search terms were verified by the two senior authors (M.K. and J.Y.), who are experienced orthopaedic surgeons. Studies were selected for inclusion if they met the following criteria: (I) included the use of AI models to aid in patient education; (II) related to shoulder and elbow orthopaedics; (III) human studies; (IV) studies published in the English language; and (V) any level of evidence I–IV. Exclusion criteria included (I) textbook chapters; (II) case reports; (III) conference abstracts or editorials; and (IV) biomechanical or cadaveric/animal studies. References of included studies and of pertinent review papers were manually searched to ensure that all means of study identification were exhausted.

Screening

Title and abstract screening were conducted by two authors (J.D.F. and M.M.) independently, with conflicts resolved through consensus or consultation with a more senior author (P.R.) if no consensus was reached. During the full-text screening stage, studies were independently screened by the initial two authors, and disagreements were resolved in a similar manner.

Assessment of agreement

Inter-reviewer agreement was evaluated using the κ-statistic for screening. A priori classification was determined using the following criteria: κ of 0.91–0.99 was considered to be almost perfect agreement; κ of 0.71–0.90 was considered to be considerable agreement; κ of 0.61–0.70 was considered to be high agreement; κ of 0.41–0.60 was considered to be moderate agreement; κ of 0.21–0.40 was considered to be fair agreement; and a κ value of 0.20 or less was considered to be no agreement (18).

Quality assessment

The Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) guidelines were used to assess the methodological quality and risk of bias in the studies used for this review (19). These guidelines assign one of three ratings (low risk, high risk, or unclear) to each item. The following factors are assessed: (I) patient selection; (II) index test; (III) reference standard; and (IV) flow and timing. Risk of bias is assessed for all four factors and applicability concerns are assessed for (I), (II), and (III), yielding seven ratings for each study.

Data extraction

Data were extracted in an electronic spreadsheet designed a priori (Google Sheets; Google LLC). Extracted data included study characteristics [e.g., author(s), year of publication, level of evidence], demographic data (e.g., sample size, patient age, sex, etc.), AI algorithm used, type of shoulder or elbow pathology assessed, clinical index scores, types of questions asked by patients, area under the curve (AUC) of the AI algorithms and accuracy, readability and quality of responses.

Outcome reporting and statistical analysis

The primary outcomes of this review were the accuracy of responses developed by various AI algorithms to a variety of orthopedic surgery-related questions as well as the AUC of the AI algorithm. The AUC is a summary measure of the performance of ML algorithms (20). Meta-analyses were conducted on studies that reported these outcomes and provided corresponding confidence intervals or standard deviation values. The outcomes were pooled and forest plots were generated with a subgroup analysis using DataParty (Hamilton, Ontario, Canada) with a DerSimonian and Laird random effects model for dichotomous variables. The I^2 test was used to assess heterogeneity of the included studies,

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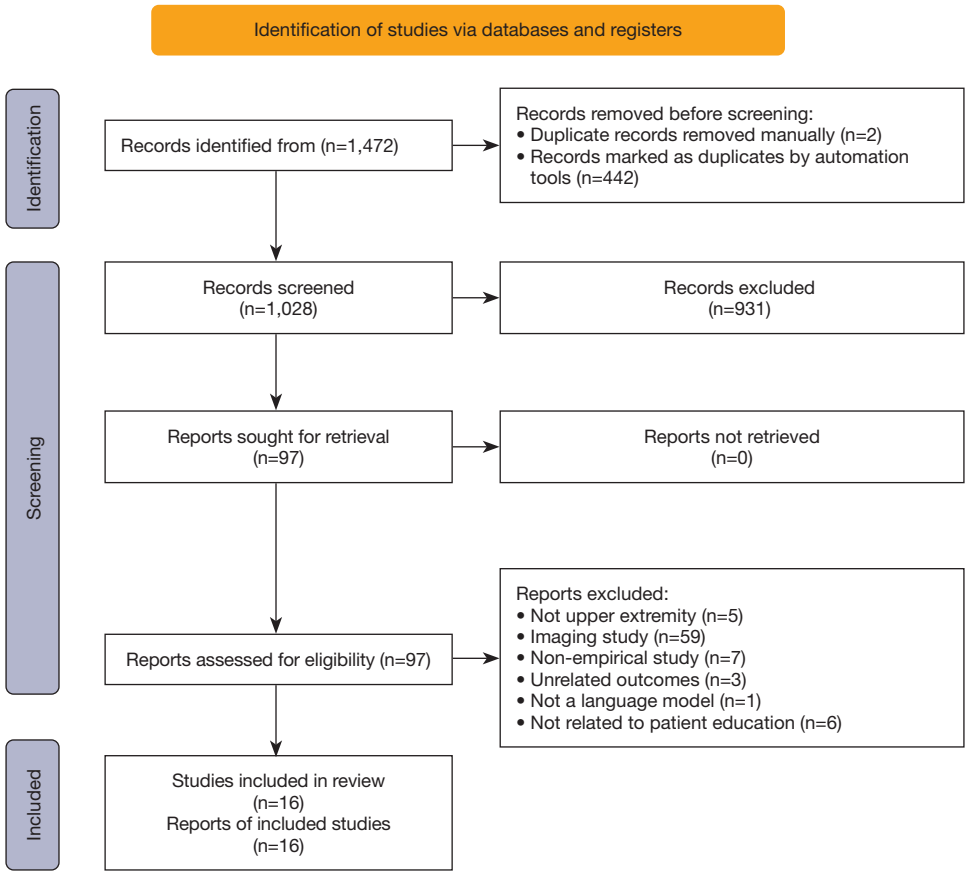


Figure 1 PRISMA flow diagram.

where "Low", "Moderate", and "High" values are identified by I^2 values of 25–49%, 50–74%, and >75%, respectively. The secondary outcomes were the quality and validity of AI responses, assessed by *Journal of the American Medical Association* (*JAMA*) benchmark score and DISCERN scores (21,22), as well as the readability, assessed by the Flesch Reading Ease Score (FRES) and Flesch-Kincaid Grade Level (FKGL) scales (23). The JAMA score ranges from 0–4 points; 0 or 1 point indicates insufficient data, 2 or 3 points indicate partially insufficient data, and 4 points indicate completely sufficient data (24). As per the DISCERN score, the quality levels are categorized as either very poor (20% to 36%), poor (37% to 50%), fair (51% to 64%), good (65% to 79%), or excellent (80% and above) (25). The FRES ranges from 0 to 100, and higher scores indicate an easier readability, whereas the FKGL score indicates the grade level that can appropriately read and comprehend a given text (26).

Results

Literature search

The initial literature search yielded 1,472 studies, of which 444 duplicates were removed. Among the remaining 1,028 unique articles, 931 were removed following title and abstract screening. Systematic screening and assessment of eligibility yielded 16 full-text studies that satisfied inclusion criteria (*Figure 1*) (7,27-41). Inter-rater reliability analysis showed that considerable agreement was achieved at both the title and abstract screening stage [k, 0.75; 95% confidence interval (CI): 0.68–0.82] and full-text (k, 0.76; 95% CI: 0.60–0.91) stages of screening.

Study quality

Among the 16 studies included in this review, 10 (62%) were level IV evidence (7,25,27-30,32,35,36,38) and 6

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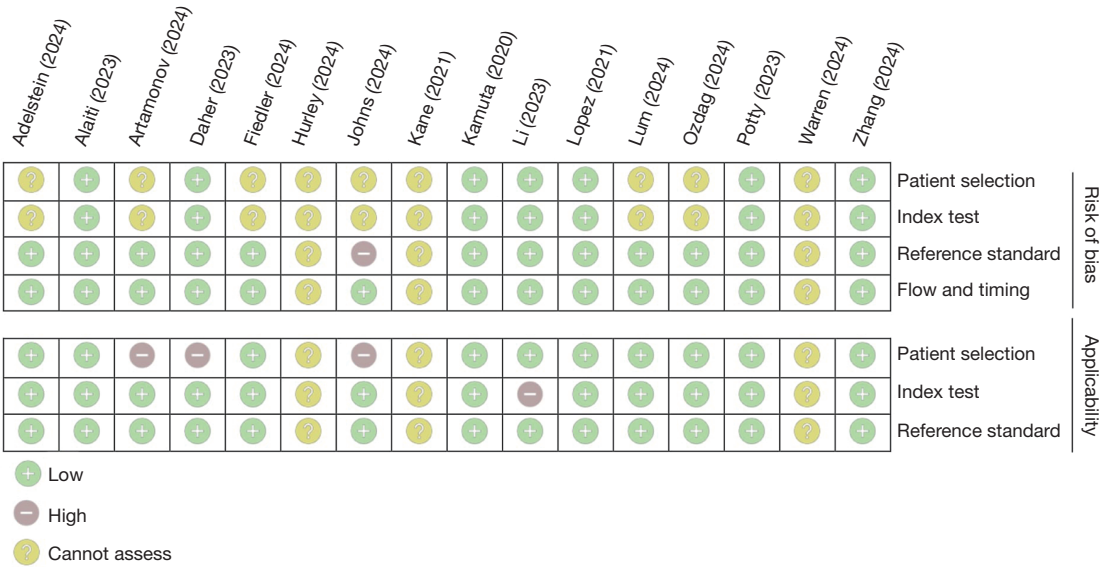


Figure 2 Quality assessment of included studies.

(37.5%) were level III evidence (26,31,33,34,37,39). All 16 studies underwent appraisal by the QUADAS-2 tool. The majority of studies were found to have an overall low risk of bias (*Figure 2*).

Study characteristics

This review included 16 studies that reported on the use of AI algorithms in answering queries related to orthopaedic surgery. A total of 117,526 patients were included across seven studies that reported values (28,30,34-36,39,41).

Regarding the AI algorithm used, nine studies used any version of ChatGPT; specifically, one (30), five (7,32,37,38,40), and two (7,29) studies used ChatGPT 3, 3.5 and 4, respectively. Two studies did not specify which version of ChatGPT was used (27,31). One study used Google Bard (37). Seven studies used a variety of other learning models (28,33-36,39,41), outlined in *Table 1*.

Characteristics of topics and questions asked

From the 16 included studies, the most commonly assessed question pertained to rotator cuff (RTC) tears or repairs, followed by shoulder arthroplasty, and upper extremity pathologies in general, as investigated in six (28,30,35,39-41), three (27,34,36), and three studies (7,37,38), respectively.

The main types of questions posed pertained to predicting clinical outcomes (four studies) (28,33,39,41), diagnostics and management planning (three studies) (29,30,35), general frequently asked questions (three studies) (27,32,40), exam writing (three studies) (7,37,38), and discharge planning (two studies) (34,36) (*Table 1*).

Performance

Fourteen studies reported on the overall mean accuracy of answers to questions asked, defined as the percent of correct responses to questions. Percentages were extrapolated when accuracy was scored using a 4-point Likert scale. Values of accuracy were found to range from 45% to 99% (7,27-30,32-39,41). Eight of these studies reported mean accuracy values of 70% or greater (28,30,32-36,41). The pooled mean accuracy of responses by AI algorithms was 78% (0.78) (95% CI: 0.4–1.16%, I^2=0%) (*Figure 3*). There was no significant difference in pooled mean accuracy between question types (P=0.99, I^2=0%) (*Figure 3*). Two studies compared the accuracy of AI responses to that of expert responses and demonstrated that experts were more accurate in both cases (7,38). One study evaluated the similarity in responses between GPT-4 and the anterior shoulder instability (ASI) consensus statement; when assessed by surgeons, similarity was most commonly classified as 'medium', and when assessed by GPT-4, similarity was most often classified as 'high' (29).

Four studies reported on the overall mean area under the receiver operating characteristic curve (AUC), with values

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Table 1 Study characteristics and patient demographics

Lead author (year of publication)	Level of evidence	Algorithm used	Number of patients	Females (%)	Mean age, years (SD)
Adelstein (2024) (27)	IV	ChatGPT (version NS)	NR	NR	NR
Alaiti (2023) (28)	III	Random forest classifier, LightGBM, decision tree classifier; extra trees classifier; logistic regression; XGBoost, KNN classifier; CatBoost classifier	474	54	57.8 (8.9)
Artamonov (2024) (29)	IV	ChatGPT 4.0	NR	NR	NR
Daher (2023) (30)	IV	ChatGPT 3	29	48.3	NR
Fiedler (2024) (7)	IV	ChatGPT 3.5, 4	NR	NR	NR
Hurley (2024) (31)	IV	ChatGPT (version NS)	NR	NR	NR
Johns (2024) (32)	IV	ChatGPT 3.5	NR	NR	NR
Kane (2021) (33)	III	CAT	NR	NR	NR
Karnuta (2020) (34)	IV	ANN	90,792	59.2	69.0 (10.9)
Li (2023) (35)	III	Stacking, GBM, bag-ging, random forest, XGBoost, adaptive boosting	1,684	53.1	49.8 (14.4)
Lopez (2021) (36)	III	BDT, ANN	21,544	55.3	69.1 (9.5)
Lum (2024) (37)	IV	ChatGPT 3.5, Google BARD	NR	NR	NR
Ozdag (2024) (38)	IV	ChatGPT 3.5	NR	NR	NR
Potty (2023) (39)	III	XGBoost	631	42.6	61.5
Warren (2024) (40)	IV	ChatGPT 3.5	NR	NR	NR
Zhang (2024) (41)	III	EV-GCN	2,372	53.9	NR

ANN, artificial neural network; BDT, boosted decision tree; CAT, computerized adaptive testing; EV-GCN, edge-variational graph convolutional network; GBM, gradient boosting machine; GPT, generative pre-trained transformer; KNN, K-nearest neighbors; NR, no result; NS, not specified; SD, standard deviation; XG, extreme gradient.

ranging from 0.72 to 0.92 (28,34-36). The pooled mean AUC of AI algorithms was 86% (0.86) (95% CI: 0.55–1.16, I^2=0%) (*Figure 4*).

Two studies reported on the overall mean quality of answers to questions asked using JAMA and DISCERN scores (31,40). From these studies, all the reported JAMA scores were 0%, and the DISCERN scores ranged from 64% to 75%. Three studies reported on the overall mean readability of answers to questions asked using the FKGL and FRES scores (31,32,40). Using FKGL as the outcome, the readability of responses was at a high school reading level in two studies and at a college graduate level in one study. FRES scores were reported in two studies; values of responses ranged from 26.2 to 48.3, representing either a college student or college graduate readability level. Full details regarding outcomes used to assess responses are shown in *Table 2*.

Discussion

The primary finding of this paper demonstrates that AI models answered questions to inquiries pertaining to shoulder and elbow orthopaedic pathologies with a pooled mean accuracy of 78%. No significant difference in accuracy was found across question types. When reported, experts answered questions with a higher level of accuracy when compared to various AI algorithms. Furthermore, this study determined that the pooled mean AUC, a holistic summary measure of AI algorithm performance (20,42), was 86%. This pooled AUC value is comparable to other AUC values of a variety of AI algorithms found in previous studies, which ranged from 80.7% to 95% (20,42-44). The secondary findings of this paper are that JAMA scores accredited to AI algorithms were all zero, indicating that the information provided by AI was insufficient or unhelpful to readers (45). Moreover, readability scores of responses

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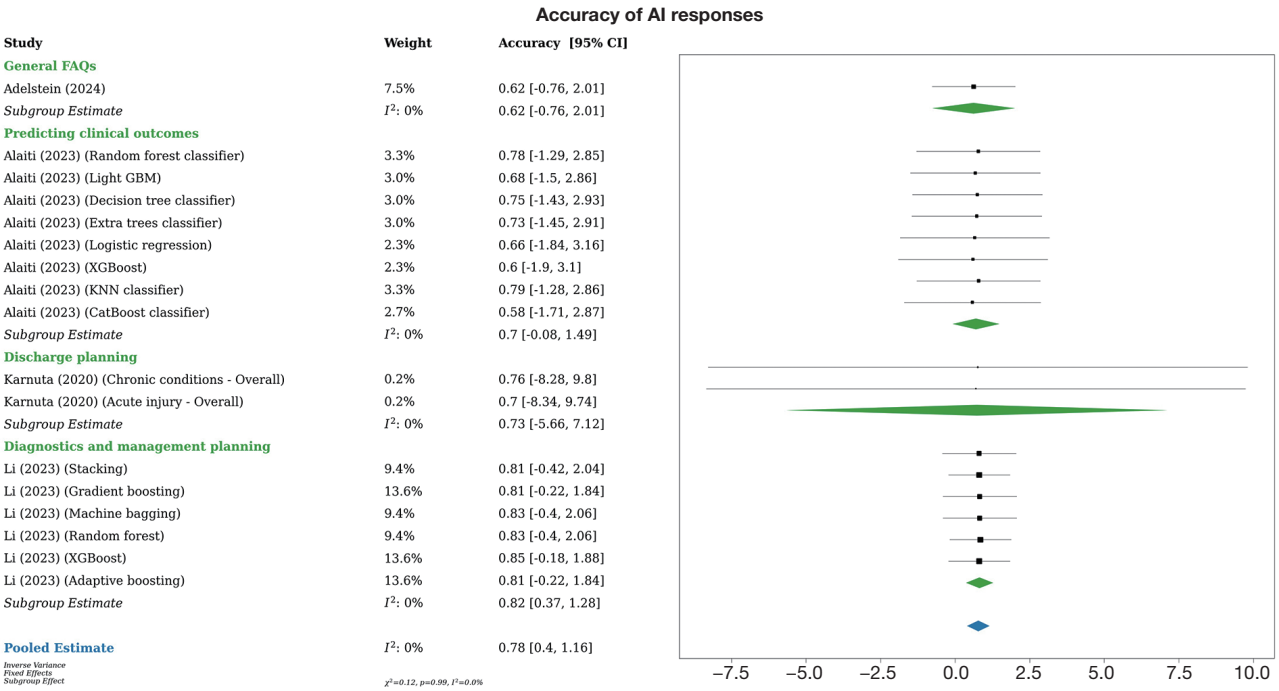


Figure 3 Forest plot of the pooled accuracies of responses developed by AI models, stratified by question type. AI, artificial intelligence; CI, confidence interval.

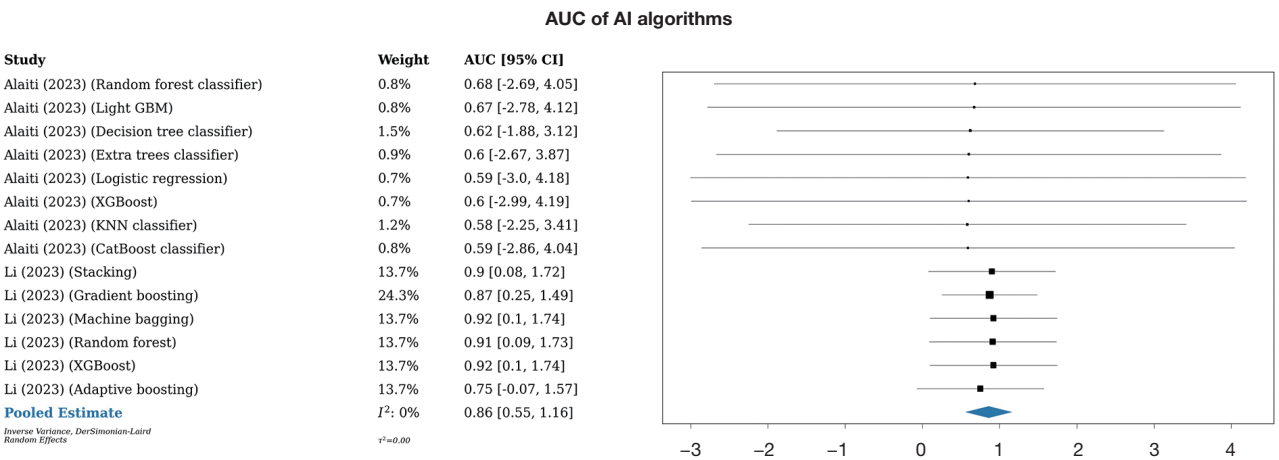


Figure 4 Forest plot of the pooled AUC of AI algorithms. AI, artificial intelligence; AUC, area under the curve; CI, confidence interval; GBM, gradient boosting machine; KNN, K-nearest neighbors; XGBoost, eXtreme gradient boosting.

developed by AI algorithms ranged from high school to college graduate reading level.

It is important to note that there was a wide range of values reported across the different studies and algorithms. While the algorithms performed well overall in terms of accuracy and AUC, the overall quality of developed responses was poor, and they performed worse than experts. The majority of patient education material is written at a level too complex, and it has been reported that most patients would find text with readability at a college level difficult to read or comprehend (46). As patient education is highly variable and specific preferences exist along a

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Table 2 Characteristics of questions asked and performance outcomes

Lead author (year of publication)	Type of upper extremity procedure/pathology	Type of question	AI accuracy (%)	Expert accuracy (%)	AUC	Quality and validity of responses	Readability of responses
Adelstein (2024) (27)	Total shoulder arthroplasty	General FAQs	62.5	NR	NR	NR	NR
Alaiti (2023) (28)	RTC repair	Predicting clinical outcomes	Random forest classifier: 78; lightGBM: 68; decision tree classifier: 75; extra trees classifier: 73; logistic regression: 66; XGBoost: 60; KNN classifier: 79; CatBoost classifier: 58	NR	Random forest classifier: 0.68; lightGBM: 0.67; decision tree classifier: 0.62; extra trees classifier: 0.6; logistic regression: 0.59; XGBoost: 0.6; KNN classifier: 0.58; CatBoost classifier: 0.59	NR	NR
Artamonov (2024) (29)	Anterior shoulder instability	Diagnostics and management planning	Overall: 58.1. Degree of similarity between GPT-4 and ASI consensus statement Assessed by surgeons: high, 25.8; medium, 45.2; low, 29 Assessed by AI: high, 48.3; medium, 41.9; low, 9.7	NR	NR	NR	NR
Daher (2023) (30)	RTC tendinopathies, glenohumeral arthritis, proximal humeral fractures	Diagnostics and management planning	Diagnosis: 93; management: 83	NR	NR	NR	NR
Fiedler (2024) (7)	General ASES questions	Exam questions	ChatGPT 3.5: text-only: 60.8; media-based: NR ChatGPT 4: text-only, 66.7; media-based, 53.2	Text-only: 76.4; media-based: 73.9	NR	NR	NR
Hurley (2024) (31)	Shoulder stabilization surgery	Details surrounding surgery	NR	NR	NR	JAMA: 0; DISCERN: 60	FRES: 26.2; FKGL: (NR), college graduate level
Johns (2024) (32)	UCL reconstruction	General FAQs	75	NR	NR	NR	FKGL: 11.51, high-school level
Kane (2021) (33)	Hand and wrist pathologies	Predicting clinical outcomes	DASH: 99; QuickDASH: 98	NR	NR	NR	NR
Karnuta (2020) (34)	Anatomic and reverse shoulder arthroplasty	Discharge planning	Chronic conditions: overall, 76.3; cost, 76.5; LOS, 91.8; patient disposition, 73.1	NR	Chronic conditions: cost, 0.75; LOS, 0.89; patient disposition: 0.77	NR	NR

Table 2 (*continued*)

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Table 2 (*continued*)

Lead author (year of publication)	Type of upper extremity procedure/pathology	Type of question	AI accuracy (%)	Expert accuracy (%)	AUC	Quality and validity of responses	Readability of responses
			Acute injury: overall, 70; cost, 70.3; LOS, 79.1; patient disposition, 72		Acute injury: cost, 0.72; LOS, 0.78; patient disposition, 0.79		
Li (2023) (35)	RTC tears	Diagnostics and management planning	Stacking: 81; gradient boosting: 81; machine bagging: 83; random forest: 83; XGBoost: 85; adaptive boosting: 81	NR	Stacking: 0.9; gradient boosting: 0.87; machine bagging: 0.92; random forest: 0.91; XGBoost: 0.92; adaptive boosting: 0.75	NR	NR
Lopez (2021) (36)	Total shoulder arthroplasty	Discharge planning	Nonhome discharge: BDT, 90.3; ANN, 89.9	NR	Nonhome discharge: BDT, 0.788; ANN, 0.851	NR	NR
			Postoperative complication: BDT: 95.5; ANN: 92.5		Postoperative complication: BDT, 0.795; ANN, 0.788		
Lum (2024) (37)	General upper extremity pathologies	Exam questions	OITE questions; ChatGPT: 54.3; BARD: 58.2	NR	NR	NR	NR
Ozdag (2024) (38)	General upper extremity pathologies	Exam questions	45	PGY1: 51; PGY5: 76	NR	NR	NR
Potty (2023) (39)	RTC repair	Predicting clinical outcomes	69	NR	NR	NR	NR
Warren (2024) (40)	RTC repair	General FAQs	NR	NR	NR	Fact: *JAMA*, 0; DISCERN, 51	Fact: FRES, 48.3; FKGL, 10.3, high-school level
						Policy: *JAMA*, 0; DISCERN, 53	Policy: FRES, 42; FKGL, 10.9, high-school level
							Value: FRES, 38.4; FKGL, 11.6, high-school level
Zhang (2024) (41)	RTC retears following repair	Predicting clinical outcomes	96.9	NR	NR	NR	NR

AI, artificial intelligence; ANN, artificial neural network; ASES, American Shoulder and Elbow Surgeons; ASI, anterior shoulder instability; AUC, area under the curve; BDT, boosted decision tree; DASH, Disabilities of the Arm, Shoulder and Hand; FAQ, frequently asked question; FKGL, Flesch-Kincaid Grade Level; FRES, Flesch Reading Ease Score; GBM, gradient boosting machine; GPT, generative pre-trained transformer; *JAMA*, *Journal of the American Medical Association*; KNN, k-nearest neighbors; LOS, length of stay; NR, no result; NS, not specified; OITE, orthopaedic in-training examination; PGY, postgraduate year; RTC, rotator cuff; UCL, ulnar collateral ligament; XG, extreme gradient.

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continuum (47), AI algorithms must be shown to effectively adapt to meet unique patient learning needs. Therefore, for patient education material to be helpful, answers must meet standards regarding accuracy, readability, and quality. Our study demonstrates that further research and development is required before patients can unilaterally rely on communicating AI models for effective education. Moreover, important insight is gained from understanding what types of questions are most valuable from a patient's perspective, which will ultimately help guide the development of AI models and their routine use in patient education. While our review assists in achieving this goal, further research should be conducted to better comprehend how to maximize the value of AI for patients.

A major strength of this study is its focus on shoulder and elbow pathology, as it provides new insight into AI algorithm performance in answering questions in this field. As AI is inevitably continuously improved, this review offers key insight into the models' real-world applicability as a tool for patient education. Having the knowledge of the types of questions patients are asking can help inform future doctor-patient conversations as well as informational material. Furthermore, this review was conducted with a strong methodology and benefits from quantitative evidence arising from meta-analyses. In addition, our systematic review includes studies that assessed a broad range of AI models, accounting for the rapid growth of AI in patient education.

The primary limitation of this study is the wide variability of the outcomes of the included studies. Many of the studies reported on inconsistent outcomes and those with similar outcomes reported a wide range of performances across their respective AI models. As such, while the pooled data show promising measures of accuracy and AUC, the inconsistencies between studies necessitate caution in applying these models. Further, variations in types of AI models used across included studies precluded quantitative analysis to compare the efficacy of each model assessed. Another limitation of our study is that our literature search was conducted from database inception until June 6th, 2024; in consideration of the rapidly developing field of AI, it is possible that novel research has been published since our analysis.

Future directions of this research study include reassessing ongoing improvements in technology as well as investigating the cost-benefit analysis of implementing AI models throughout the healthcare system. It will be crucial to evaluate the effectiveness of AI models in patient education once integrated into authentic clinical environments.

While our study does not fully support replacing experts with AI for guiding patient education, the results show promise for its implementation in the future. As shown in our study, AI models may provide accurate answers and perform with a high AUC; however, there are several limitations for their effective application in patient education. There are already several disciplines across healthcare in which AI performs as well as or even better than clinicians (48-51). For example, if AI models consistently diagnose orthopaedic pathologies more quickly and accurately than experts, overall spending and substantial resources can then be diverted towards improving medical education and clinical training (9). Our analysis is in agreement with other literature, which suggests that, based on our current technology, AI algorithms in isolation do not perform well enough to replace experts; however, performance is increased when expert clinicians use AI as an augment. Thus, orthopaedic surgeons may advise patients to use AI models to guide understanding of simple orthopaedic problems, however, they should not rely on AI responses for definitive answers. Shared decision-making and communication between the patient and surgical team are still necessary to ensure high-quality patient education. The gaps that have been identified in our study should be addressed before AI models are fully integrated into patient education.

Conclusions

In conclusion, the performance of AI models in answering questions related to shoulder and elbow orthopaedic pathologies reflected a high degree of accuracy. However, AI responses were found to be inferior to experts and were poor in readability, quality, and value to the patient. AI is a powerful tool that can be used in conjunction with experts to augment patient education, however it cannot yet be utilized independently.

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Footnote

Reporting Checklist: The authors have completed the PRISMA reporting checklist. Available at https://aoj.

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Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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